

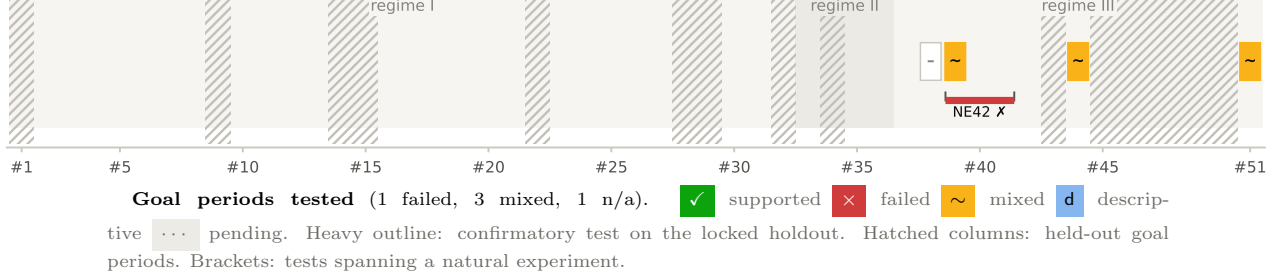
H119: The room merge switches a known adjacency on and off: recover it from the dynamics (NE42)

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1.5/5 Usefulness 2/5

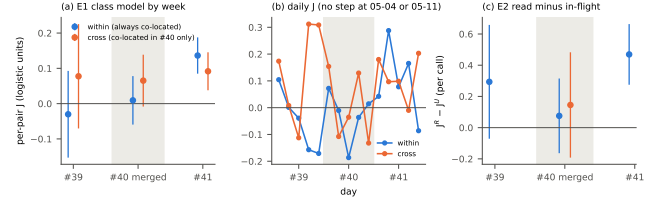
Bottom line. The NE42 merge did not switch on a coupling between the two groups; per-pair talk coupling vanished for all pairs in the merged room and returned within the old rooms after the split, which is attention dilution, not adjacency gating.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH359 says that NE42's merge, one week in which #best and #rest shared one room, switches the read path between the two groups on and off. A kinetic Ising coupling fitted from talk timing should follow it. It should be near 0 across the wall, positive in the merged week, back near 0 after the split, and carried by read-outs. H119 fits per-pair class couplings on minute talk spins for 11 agents over #39, #40 and #41. It adds a call-clock contrast of read against in-flight messages and four other room switches. **The cross coupling does not rise:** the merge contrast is $M = -0.02 [-0.13, 0.09]$ per pair, against a synthetic power of 0.95 at 0.3. Within-room pairs lose their coupling in the merged week as well, and regain it after the split (read minus in-flight 0.47 [0.27, 0.66]). HH359's kill rule fires. The holdout is not run.



Observables. (a) Per-pair class couplings by week (95% CI); grey band: the merged week. Cross pairs (orange) never rise, and within pairs (blue) are near 0 until #41. (b) Daily fits show no step on 05-04 or 05-11. (c) Read minus in-flight on the call clock: no class couples in the merged week; within pairs couple after the split.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: merging two groups into one room did not make them couple. For a week it removed measurable talk coupling for everyone: read-gated coupling of old room-mates went from 0.29 to 0.08 to 0.47 across #39, #40 and #41. To link two groups, keep rooms small and move or post across instead of merging. A read path is necessary for coupling but not sufficient: in a 14-agent room the per-pair coupling falls below what five 4-hour days resolve.

PHYSICAL MODEL

Model 02, kinetic Ising on talk spins with a block-structured J . Each pair class c (within, cross) shares one per-pair coupling J_c , and recipient fields per (day, 30-min block) absorb the schedule and the goal. Adjacency gating predicts $J_x \approx 0$ when the groups sit in different rooms and $J_x \approx J_w$ when they share one. Attention dilution predicts that per-pair coupling falls as the room grows. On the call clock, a read-gated coupling raises talk after messages read just before a call (J^R), not after messages posted during it and not yet read (J^U).

$$\text{logit } P(s_i^{t+1} = 1) = h_i + \delta_i(b) + K_i s_i^t + K'_i m_i^t + \sum_c J_c \sum_{j \in c(i)} m_j^t$$

$$M = J_x^{(40)} - \frac{1}{2}(J_x^{(39)} + J_x^{(41)}), \quad R_m = J_x^{(41)} - J_x^{(39)}$$

$$C_{RU} = J^R - J^U \quad (\text{read } [t_c - L, t_c] \text{ vs. unread } [t_c, t_c + L])$$

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RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic power and size	M power 0.95; C_{RU} power 0.73, supported size 0.00	
P1 $M > 0$ (merge rise)	-0.019 $[-0.127, 0.088]$; relabel p	failed
	0.43	
P2 no remanence	R_m 0.014 $[-0.14, 0.17]$	trivial
P3 no partition memory	0.056 $[-0.010, 0.122]$	trivial
P5 daily step at merge	merged 0.002 vs 0.126 (p 0.95)	failed
P6 read-gated cross in #40	C_{RU} 0.15 $[-0.19, 0.48]$	failed
P7 switch contrast 0	$S > 2/4$ significant (#focus); 4/4 sign	partly
HH kill	M CI includes 0	fires

GOAL PERIODS IN DETAIL

G39 $\sim$ mixed
G44 $\sim$ mixed
G51 $\sim$ mixed
NE42 $\times$ failed
G38 $-$ n/a

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	0
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

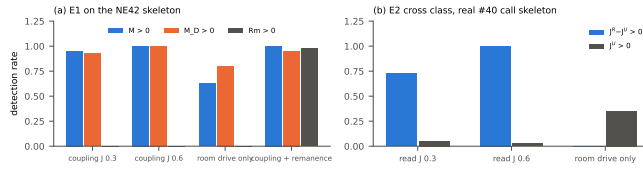
11 eligible agents. GPT-5 (the mirror arm) and most movers drop out by the 10-talk-minute rule. The minute estimator resolves per-pair couplings of about 0.2 and above. The absolute J^R and J^U are confounded by call length; only their difference is read. The minute estimator's switch contrast is not specific (room-drive false positives 0.58–0.95), and the read-gated attribution holds at one of five switches. #40's shared objective coincides with the merge, but coupling fell.

ROUND 2

Fit coupling against room size across all one- and two-room weeks (a dilution exponent). Pre-register the dilution null for the held-out #48–#49 merge.

- **Essence.** does a read path make coupling, or does attention per partner? Fit J against room size across all two-room and one-room weeks (dilution exponent, H18) instead of an on/off switch.

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Second figure. Synthetic worlds on the real NE42 skeleton (40 each). (a) The minute estimator detects a gated coupling (M) and remanence, but a shared room drive alone also gives $M > 0$ in 0.63 of worlds. (b) The call-clock read-minus-in-flight contrast detects coupling and stays silent under a room drive, although J^U alone is positive.